PiLens

network forensics for the Raspberry Pi



The friendly field guide

Turn a Raspberry Pi into a Wi-Fi test network you can actually *see* inside.

➜ TL;DR

This guide is built for skimming. Every section opens with a **TL;DR**. Boxes in the margins and colour-coded callouts carry the important bits. **Read the box, skip the prose, come back if you need it.** You will not miss anything vital by jumping around — that is the point.

Version 0.3 • AGPL-3.0-or-later • For inspecting devices and networks you are authorised to inspect. Nothing else.

How to read this (30 seconds)

➜ **TL;DR**	The one-line version. If you read nothing else, read these.
☞ **Heads-up**	Something that will bite you if you ignore it.
☛ **Pro tip**	A shortcut or nicety. Safe to skip.
✍ **Honest truth**	Where PiLens tells you what it *cannot* do. Read these — they save hours of confusion.

> ☛ **Pro tip**
>
> In a hurry? Jump straight to **§1 The ten-minute start**, connect a phone, and open **Live Capture**. Come back for the rest when you are curious.

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1 The ten-minute start

➜ TL;DR

Flash the SD card > plug in **Ethernet** > run one installer > open the web page > finish the wizard > connect a test device to the new Wi-Fi > watch its traffic. That is the whole dance.

What you need

- A Raspberry Pi **4 or 5** running Raspberry Pi OS Lite (64-bit).
- An **Ethernet** cable to your normal network.
- The Pi's built-in Wi-Fi (this becomes the test network).
- A phone or laptop you own, to be the "device under test".

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The steps

1. Copy the release onto the Pi and run `sudo ./install.sh`. It explains everything before it touches anything, backs up your config, and **rolls back if anything fails**.
2. It prints three things: a **web address**, a **certificate fingerprint**, and a one-time **setup token**. Keep them.
3. Open the web address from a computer on your wired network. Your browser will warn about the certificate — that is expected (see §2). Check the fingerprint matches, then continue.
4. Work through the **setup wizard**: create your login, choose the management network, name the Wi-Fi, pick a password, done.
5. Connect your test device to the new Wi-Fi network.
6. Open **Live Capture**, press **Start capture** on the Wi-Fi interface, and watch the packets roll in.

☞ Heads-up

The moment you create the access point, the Pi's own Wi-Fi *becomes* that access point. If you were connected to the Pi over Wi-Fi you will be kicked off. **Always manage it over Ethernet.**

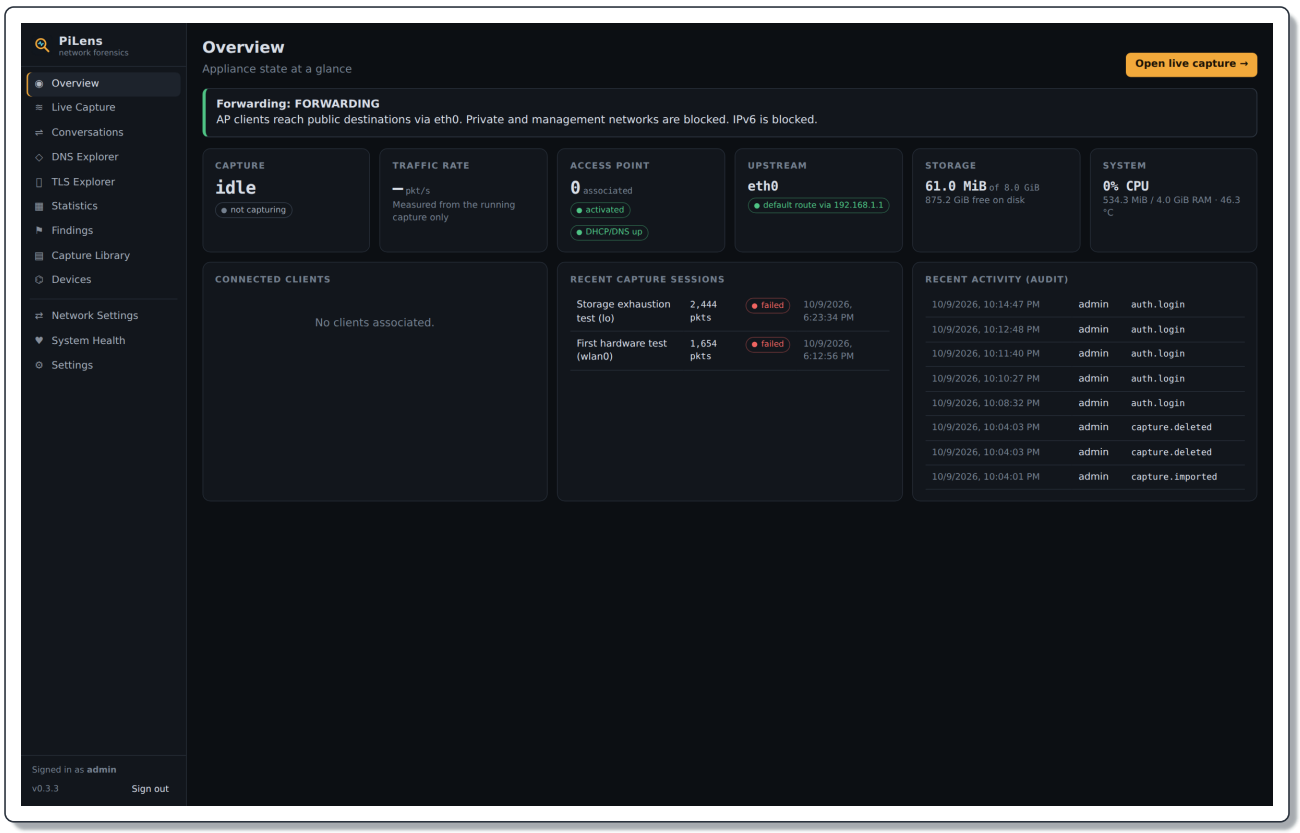


Figure: The Overview dashboard — appliance status, forwarding state, clients, and a live glance at everything at once.

2 What PiLens is (and isn't)

➜ TL;DR

PiLens is a **router with X-ray vision**. Devices on its Wi-Fi go through it to reach the internet, and it records every packet. It shows you *metadata and packets* — it does **not** decrypt HTTPS.

PiLens sits between a test device and the internet. The device joins its Wi-Fi, gets an address, and routes out through the Pi's Ethernet. Because *every* packet crosses the Pi, PiLens can capture and analyse all of it — addresses, ports, DNS, TLS handshakes, timing, the lot.

✍ The honest truth

Encrypted means encrypted. HTTPS, QUIC, VPNs and encrypted DNS stay scrambled. PiLens shows you *who* talked to *whom*, *when*, over *what* — the server name, the TLS version, the certificate when it is sent in the clear — but **not the contents**. That is TLS doing its job, not PiLens failing. Decryption needs a separate, explicit step on each device (a future feature), and **a trusted certificate can never be pushed down the network** — the device's owner must install it by hand. Anything that claimed otherwise would be lying to you.

The two padlocks

- **The management padlock** is the HTTPS on the web interface. It is self-signed, so your browser warns once. Check the fingerprint the installer printed, trust it, move on.
- **The interception padlock** (decrypting a device's HTTPS) is a *separate*, opt-in, per-device thing that is not part of this version.

3 A tour of the screens

➜ TL;DR

Left-hand menu, top to bottom: **Overview** (glance), **Live Capture** (the main event), **Conversations** / **DNS** / **TLS** / **Statistics** / **Findings** (ways to slice what you caught), then **Library, Devices, Network, Health, Settings** (housekeeping).

Live Capture — the main event

➜ TL;DR

Three stacked panes, exactly like Wireshark: the **packet list** up top, the **protocol tree** in the middle, the **raw bytes** at the bottom. Click a packet; the other two follow.

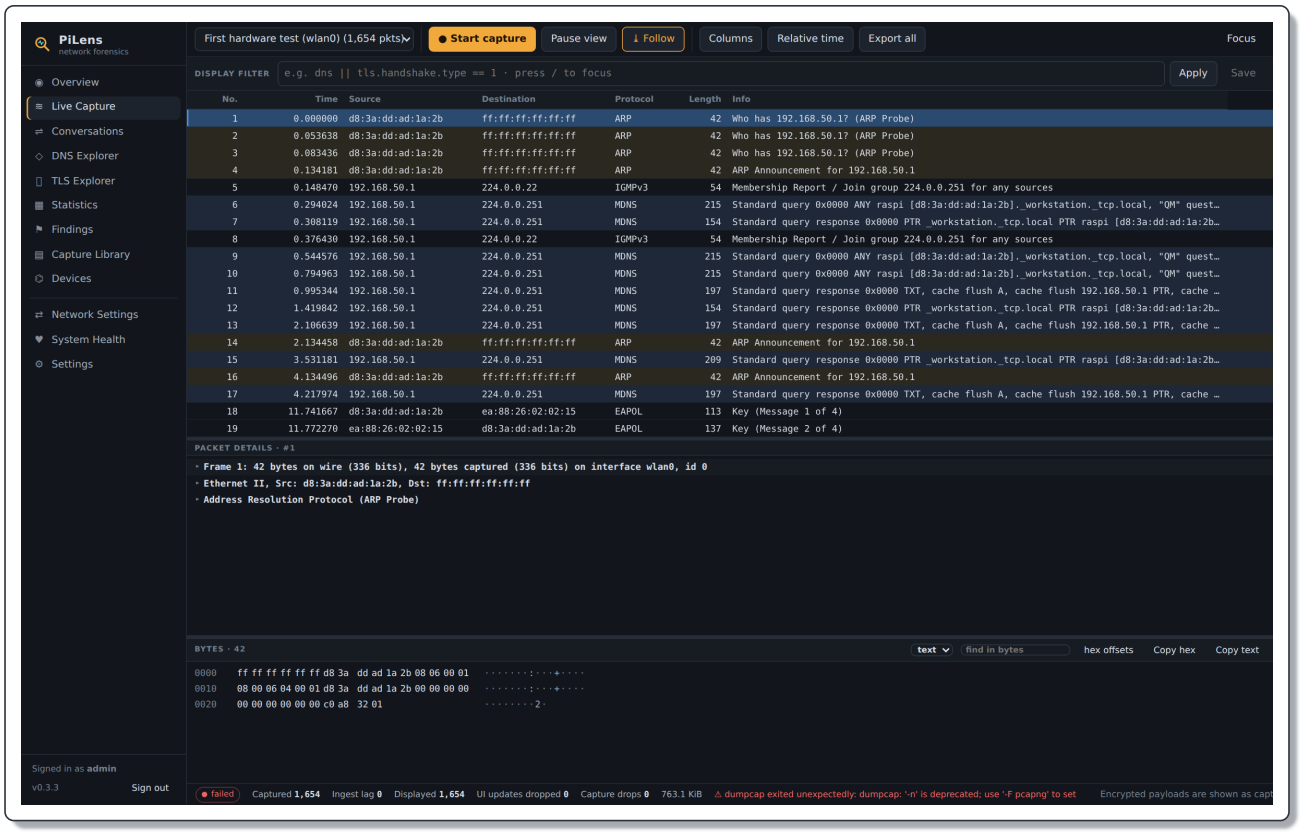


Figure: Live Capture. Packet list, protocol tree and a hex/ASCII inspector. The status bar along the bottom keeps you honest.

The packet list streams live and is colour-coded by protocol. It stays smooth even on a busy network because only the visible rows are drawn.

The protocol tree shows the selected packet decoded field by field, by TShark (the Wireshark engine). Click a field and the matching bytes light up below — but *only* when PiLens can prove those bytes really are that field. It will never make up a highlight.

The byte pane is a proper hex inspector: offsets, hex, ASCII, search, copy-as-hex or text.

☛ Pro tip

The bottom status bar separates things people usually confuse: **Captured** (packets stored), **Ingest lag** (stored but not yet summarised), **UI updates dropped** (your browser fell behind — harmless), and **Capture drops** (the appliance genuinely missed packets — not harmless). A dropped *screen update* is not a dropped *packet*.

☞ Heads-up

Pause view and **Pause capture** are different buttons. “Pause view” just freezes the scrolling so you can read; capture keeps running. “Pause capture” actually stops storing packets. “Stop” ends the session for good.

Filters — find the needle

➜ TL;DR

Type a **Wireshark display filter** in the bar (e.g. `dns`, `tcp.port == 443`, `ip.addr == 192.168.50.20`) and press Enter. It is evaluated by TShark, so real Wireshark syntax works.

The box turns green when the filter is valid, red when it isn’t, with the reason. Press `/` anywhere to jump to it. Right-click a packet or a field to build a filter without typing.

Conversations & Follow TCP Stream

➜ TL;DR

Conversations groups traffic by endpoint pair with bytes each way. Hit **Follow** on a TCP row to reassemble the whole back-and-forth.

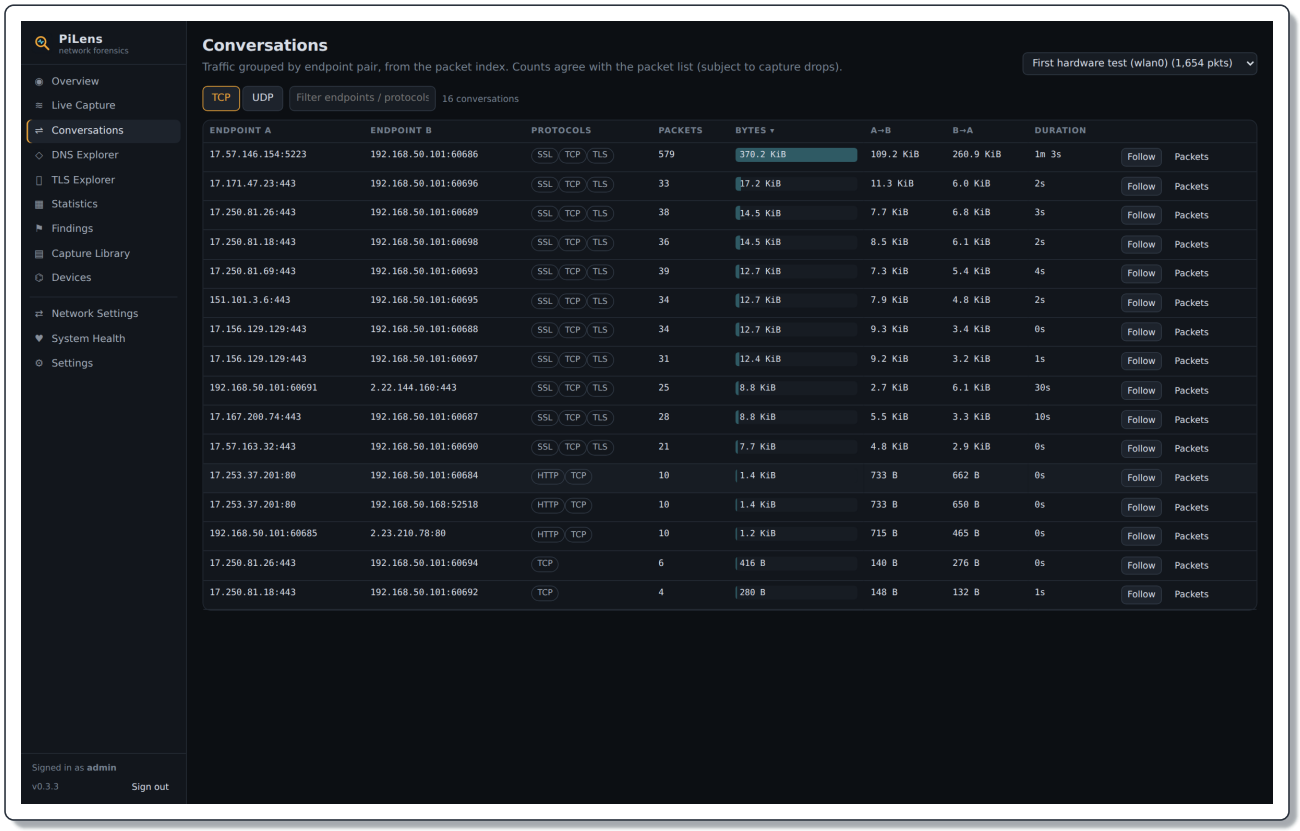


Figure: Conversations — who spoke to whom, how much each way, sortable, with one-click Follow and filtering.

✍ The honest truth

“Follow” shows the *transport stream* — the bytes as they crossed the wire. For an HTTPS connection that is **ciphertext**, and PiLens labels it as such. It is honest about the difference between reassembled packets and decrypted content.

DNS Explorer

➜ TL;DR

Every cleartext name lookup: who asked, for what, what came back, and whether it failed. Group by name, or filter to errors only.

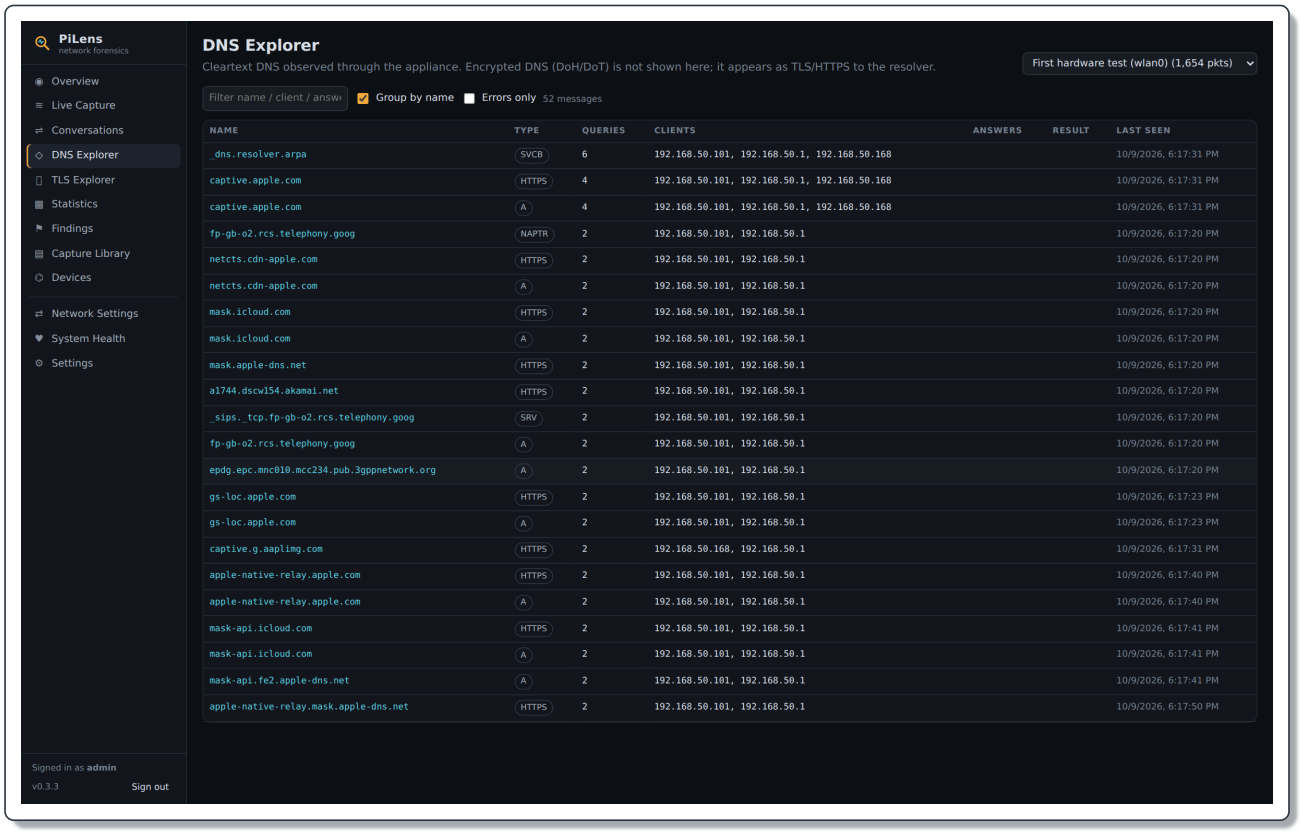


Figure: DNS Explorer, grouped by name — queries, answers, record types and result codes.

☞ Heads-up

If a device uses **encrypted DNS** (DoH/DoT), its lookups are inside TLS and *won't* appear here — they look like HTTPS to a resolver. PiLens never pretends encrypted DNS is visible.

TLS Explorer

➜ TL;DR

Every TLS/QUIC handshake: server name (SNI), negotiated version, cipher, ALPN, and any certificate sent in the clear. Metadata only — marked "encrypted pass-through" on every row.

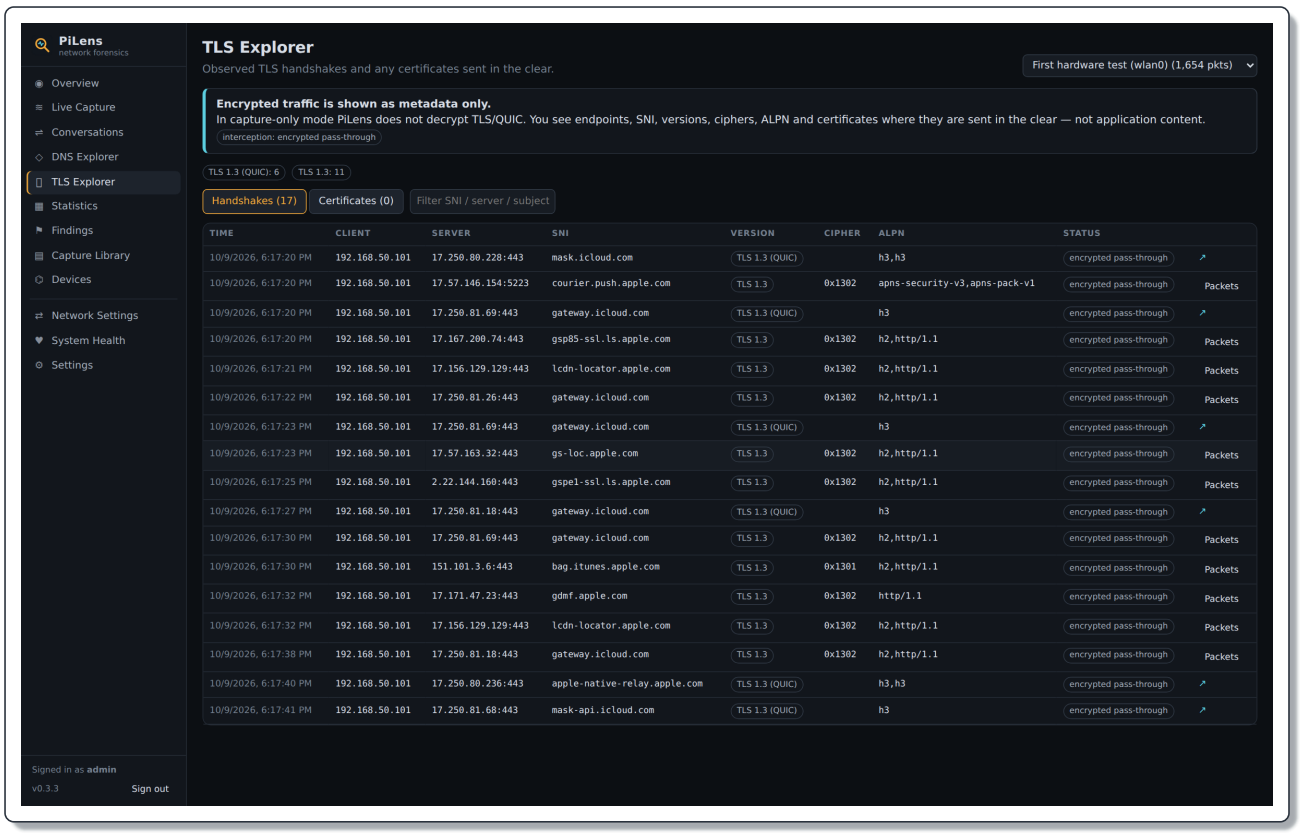


Figure: TLS Explorer. Real server names and versions, honestly labelled. The certificate tab is usually empty on modern servers — TLS 1.3 encrypts the certificate.

☛ Pro tip

The version column reports the *true negotiated* version, not the compatibility value on the wire. QUIC shows as “TLS 1.3 (QUIC)”. If you ever see everything claiming “TLS 1.2”, that’s a bug in a lesser tool — here it’s read from the handshake properly.

Statistics

➜ TL;DR

The shape of the capture: traffic over time, protocol mix, top talkers, packet sizes. Toggle bytes vs packets. Every number is a real count, defined in the panel on the right.

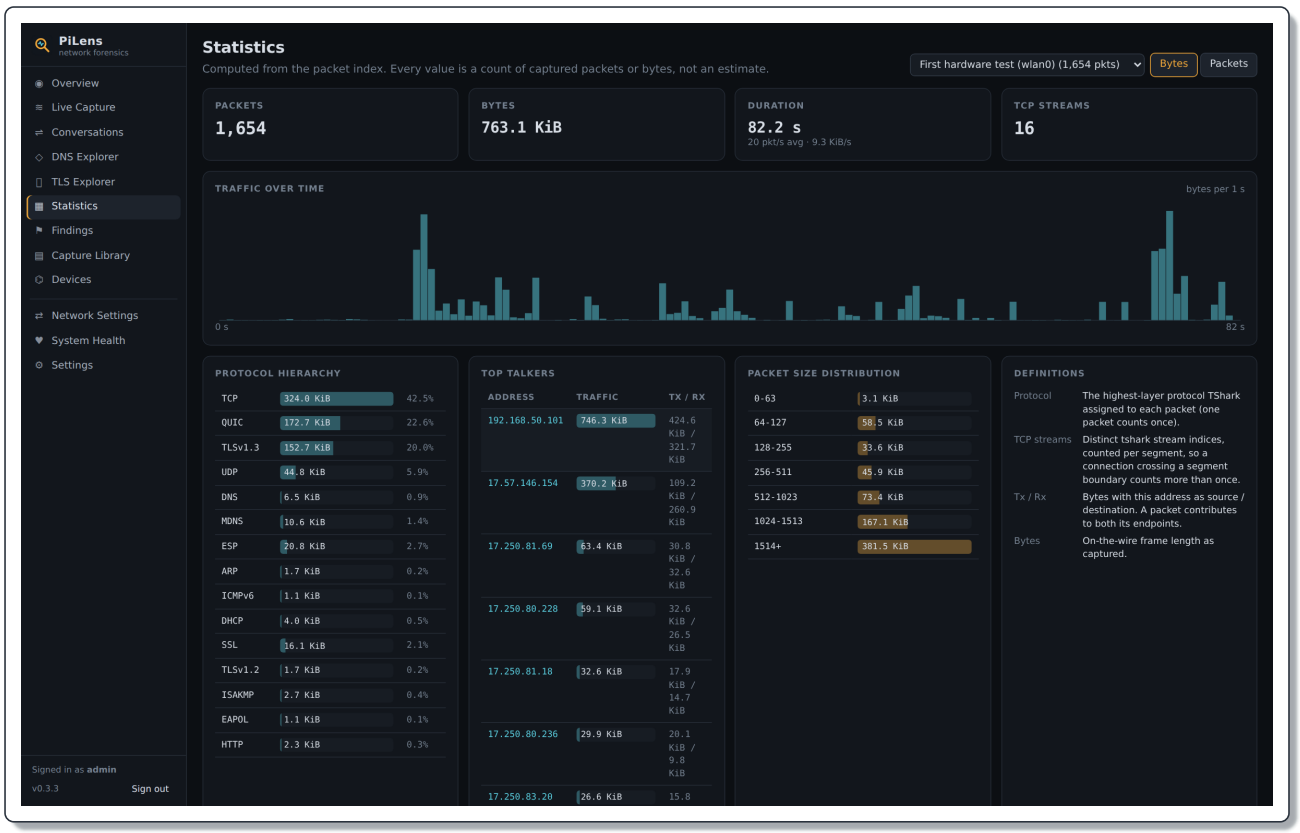


Figure: Statistics — a traffic histogram, protocol hierarchy, top talkers and size distribution, with a definitions panel so nothing is a mystery.

Findings

➜ TL;DR

Observable hiccups — retransmissions, resets, SERVFAILs, NXDOMAINs, dropped packets — each with evidence you can click straight to.

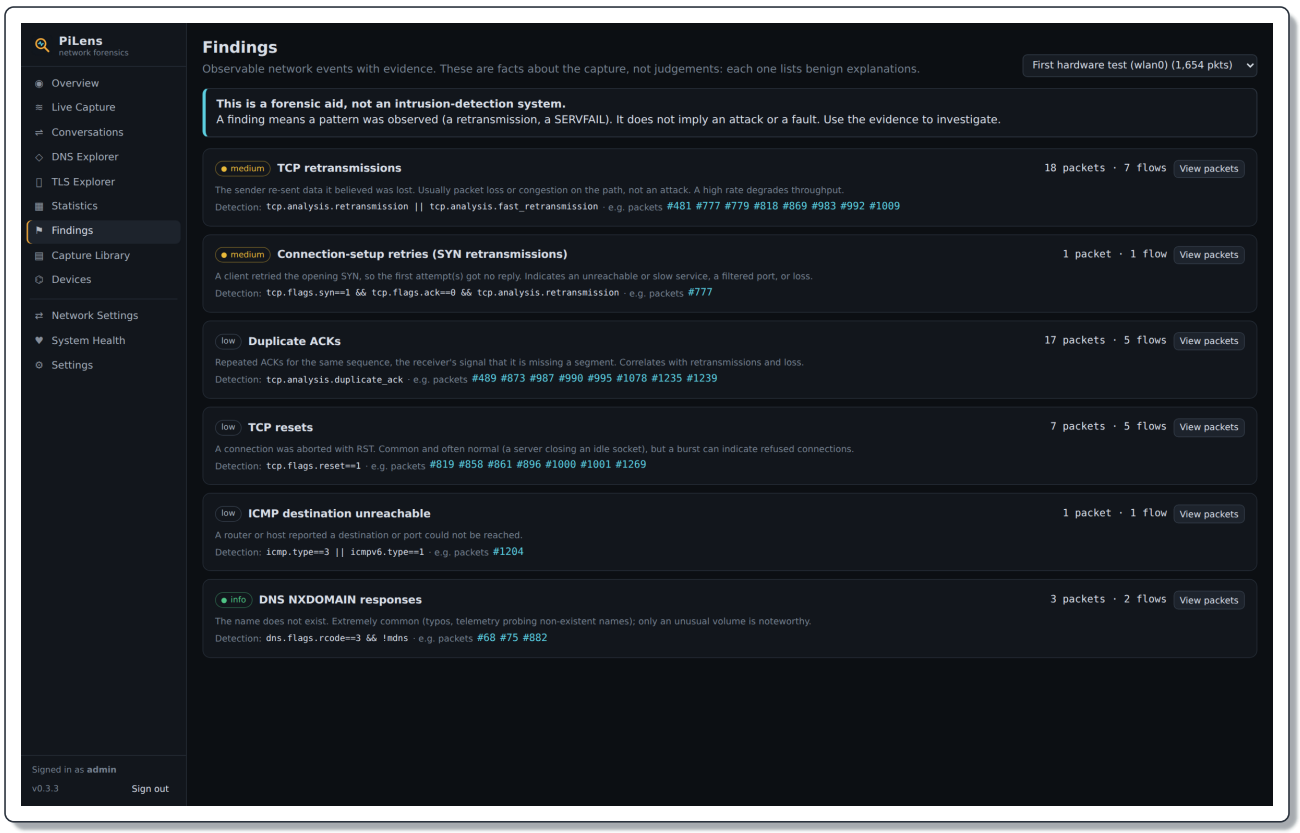


Figure: Findings. Each card is a fact with example packets, a reproducing filter, and a note on benign causes.

✍ The honest truth

Findings is a **magnifying glass, not a smoke alarm**. A finding means a pattern was *observed*, not that something is wrong or malicious. Each card spells out the innocent explanations. PiLens will never tell you “you’ve been hacked” from a retransmission.

The housekeeping screens

Capture Library

Every saved session. Rename, tag, add notes, mark *preserved* (never auto-deleted), export, or check integrity (SHA-256 per file + a manifest).

Devices

Who has been on the Wi-Fi. Label them, add notes, jump to their packets.

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Network Settings

Change the access point, DHCP, DNS, forwarding and management access. Applied as one transaction with automatic rollback.

System Health

CPU, memory, temperature, services, storage, interfaces — each with what was actually measured.

Settings

Theme, density, password, two-factor, storage limits, colour rules, and the audit log.

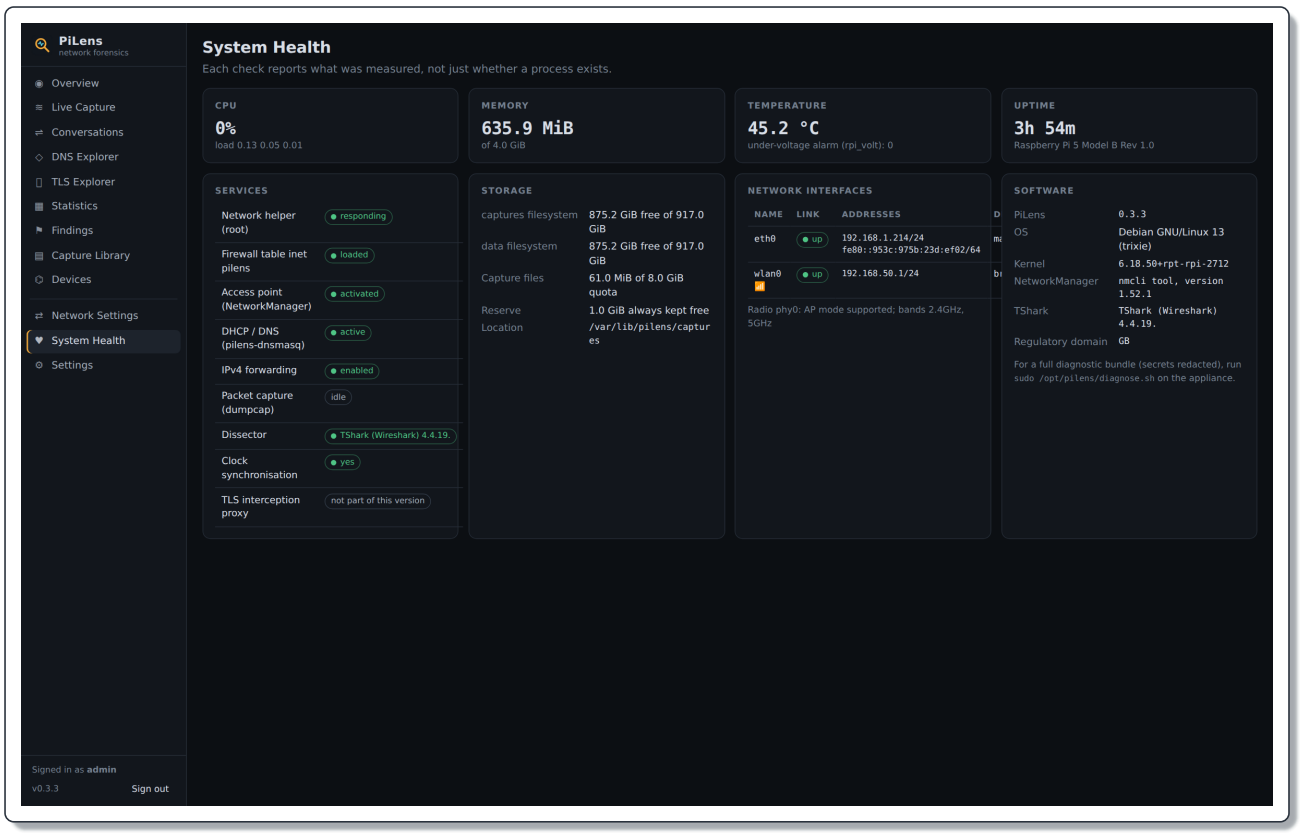


Figure: System Health — every service reports what it actually did, not just whether a process exists.

4 Common tasks, as recipes

➜ TL;DR

Short how-tos for the things you'll actually do. Find the one you need, ignore the rest.

"See everything one device does"

1. **Devices** > find it > **Packets**. (Or label it first so you recognise it.)
2. You land in Live Capture filtered to that device. Add `&& dns` or `&& tls` to narrow further.

"Save this capture as evidence"

1. Stop the capture. **Library** > the session > **Edit** > tick **Preserved**, add notes.
2. **Integrity** > **Verify hashes**, then **Download manifest**. The original `.pcapng` opens in desktop Wireshark too.

"Analyse a capture from somewhere else"

1. **Library** > **Import pcap/pcapng**. It indexes in the background; the file is kept exactly as you uploaded it.

"Stop a device reaching the internet but still watch it"

1. **Network Settings** > **Forwarding** > **No forwarding** > apply. Clients still get an address and DNS, so you can see what they *try* to reach.

5 When something's off

➜ TL;DR

Run `sudo /opt/pilens/diagnose.sh` first (it redacts secrets). Then the table.

Symptom	Try this
Browser cert warning	Expected. Match the fingerprint, trust it.
Can't log in	`sudo -u pilens /opt/pilens/bin/pilens reset-password -user admin`
Locked out by the subnet	On the console: `sudo nft delete table inet pilens`, fix it in the UI, re-apply.
No internet for clients	Check the Overview **Forwarding** card and your Ethernet uplink.
AP won't start	Try WPA2 / 2.4 GHz / a legal channel. Failed applies roll back by themselves.
Capture shows drops	Use a capture filter; record to an SSD, not the SD card.
Everything's wonky	`sudo /opt/pilens/repair.sh` fixes permissions and restarts services.

☞ Heads-up

If a network change ever strands you, every apply keeps a backup and rolls back on failure, and the last-good firewall is restored at boot before the Wi-Fi comes up. You are rarely more than a reboot from a working appliance.

6 Keyboard cheat sheet

`/`	focus the filter	`Space`	pause / resume the view
`↑` `↓`	move through packets	`f`	follow newest packet
`PgUp` `PgDn`	jump a page	`Shift+F`	focus mode (hide chrome)
`←` `→`	collapse / expand tree	`Alt+1..7`	jump between pages
`Home` `End`	first / last packet	right-click	build a filter

✍ The honest truth

The one rule that matters: only point PiLens at devices and networks you own or are explicitly allowed to inspect. It has no hidden, covert or impersonation features, on purpose. Captured traffic can hold passwords and private messages — treat your captures like the sensitive data they are, and delete what you don't need. Access and exports are logged.

PiLens — built to be fast, honest, and genuinely useful.
Full docs live in the repository's `docs/` folder.